

Project Title:

Stroke Risk Analysis Dashboard

Project Overview

This project analyzes clinical and lifestyle factors associated with stroke risk using a healthcare dataset containing **5,110 patient records**. The objective was to perform exploratory data analysis and develop an **interactive Excel dashboard** to identify patterns related to stroke occurrence across demographic and health indicators.

Tools & Techniques Used

- Microsoft Excel
 - Data cleaning and transformation
 - Feature engineering (Age Groups, BMI Categories)
 - Pivot Tables and Pivot Charts
 - Dashboard design with KPIs and slicer-based filters
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Key Metrics

- **Total Patients:** 5,110
 - **Total Stroke Cases:** 249
 - **Overall Stroke Rate:** 4.9%
 - **Average Patient Age:** 43.2 years
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Dashboard Features

The interactive dashboard visualizes stroke risk across multiple health and demographic dimensions:

- Age Groups
- BMI Categories
- Smoking Status
- Gender Distribution
- Risk Factor Classification

Stroke probability was calculated using the **average of the binary stroke indicator (0 = no stroke, 1 = stroke)** to estimate risk percentages for each subgroup.

Interactive **slicers** allow users to dynamically filter results and explore how stroke risk varies across different population segments.

Key Insights

- **Age is the strongest risk indicator**, with patients aged **60+** showing the **highest stroke probability**.
 - **Former smokers exhibit the highest stroke risk**, suggesting potential long-term vascular effects of smoking.
 - **Overweight and obese BMI groups show elevated stroke probability** compared with individuals in the normal BMI category.
 - **Gender differences in stroke probability are minimal**, indicating that gender alone may not be a dominant predictor of stroke within this dataset.
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Conclusion

This project demonstrates how healthcare datasets can be transformed into **clear, interactive decision-support dashboards** using Excel. The analysis highlights key demographic and lifestyle factors associated with stroke risk and illustrates how simple analytical tools can provide valuable insights for healthcare data exploration.

Recommendations

- **Early risk identification systems** can be developed to flag high-risk patients (e.g., age 60+, smokers, high BMI) and **send alerts to healthcare providers** for timely intervention.
- Targeted **lifestyle intervention programs** focusing on smoking cessation, weight management, and preventive care can help reduce stroke risk in vulnerable groups.